# From World Formation to Framing: Boundary, Quotient, and the Construction of Operational Worlds

## A General Framing Kernel for Bounded Observers, with PORE as a Testable Low-Dimensional Specialization


### Abstract

A bounded observer never receives reality already divided into the variables, subsystems, objects, and causal categories required for reasoning. Before prediction, control, or interpretation can begin, some distinctions must be retained while others are ignored, some degrees of freedom must be treated as internal while others are treated as environmental, and some states must be admitted while others are rejected. These operations are often conflated under the broad language of "framing" or "boundary formation." This article separates them and develops a minimal operational framework for studying their interaction.

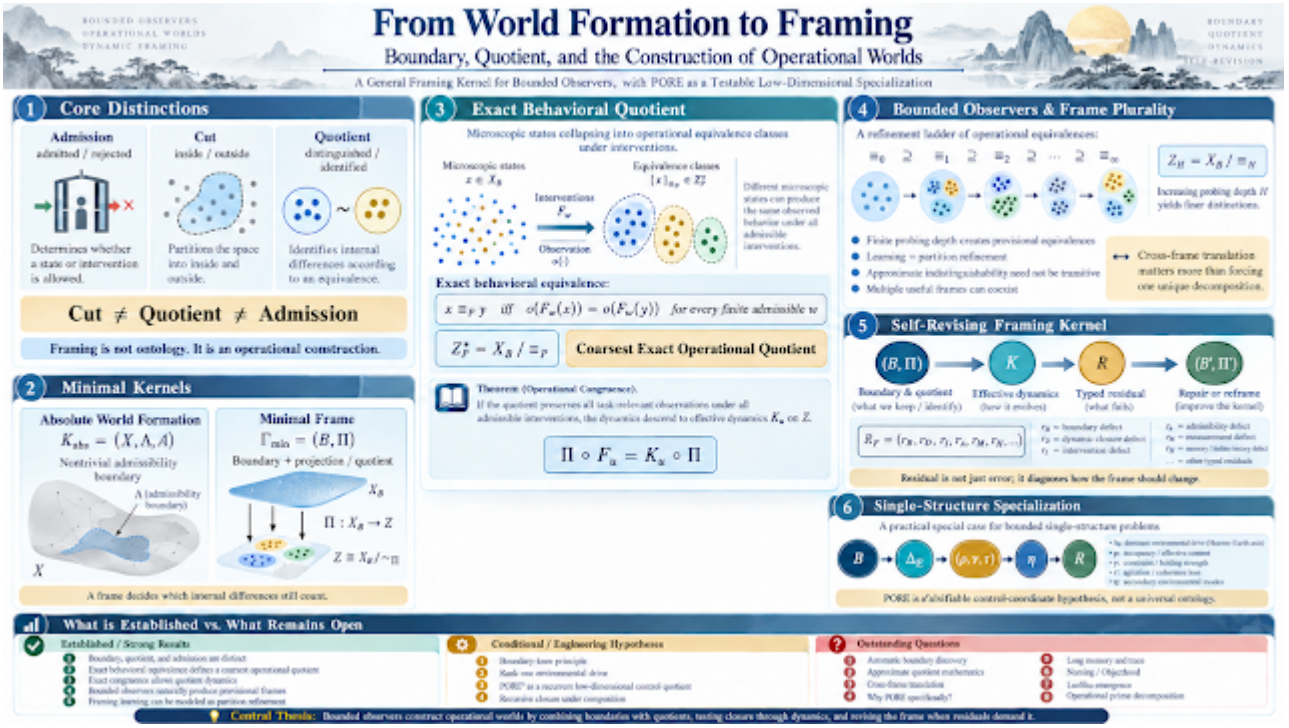
The starting point is World Formation: the formation of a nontrivial distinction or admissibility boundary. Framing is then treated as a higher-level construction rather than as an ontology. Its minimal kernel is proposed to be a pair $\Gamma_{min} = (B, \Pi)$, where $B$ specifies an operational inside–outside cut and $\Pi$ maps the retained domain into an effective quotient space. Two microscopic states are identified whenever the current frame assigns them the same effective state. Under suitable congruence conditions, the original dynamics descend to well-defined effective dynamics on this quotient.

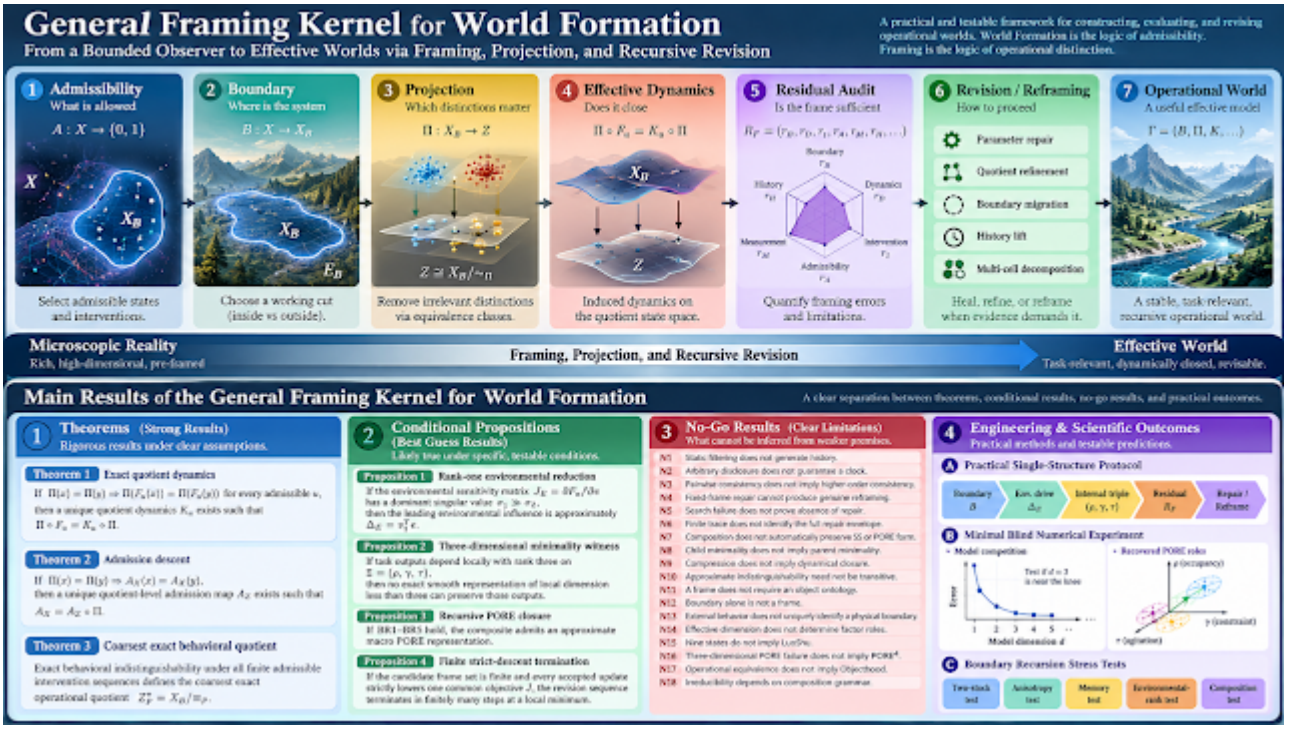
For a fixed boundary, task, observation map, and set of admissible interventions, an exact behavioral equivalence relation can be defined: two states are equivalent if no finite admissible intervention sequence can distinguish them through task-relevant observations. The corresponding quotient is the coarsest exact operational frame for that protocol. This uniqueness, however, disappears in the practical regime of bounded observers. Finite probing depth, limited intervention repertoires, noisy measurements, tolerance thresholds, and finite computational budgets generate provisional equivalences and multiple admissible frames. Learning can then be represented as partition refinement, while deeper reframing may alter either the quotient, the boundary, or both.

The framework does not assume that reality possesses one privileged decomposition. Multiple frames may remain simultaneously useful, provided that their predictions, interventions, and translations remain operationally adequate. Within this general framework, a more specific hypothesis is introduced for bounded single-structure systems. A dominant environmental drive may sometimes be extracted as a rank-one forcing mode, while the internal state may admit a coarse control representation $\Xi = (\rho, \gamma, \tau)$, representing effective occupancy, constraint or holding strength, and agitation or coherence loss. This PORE triple is treated as a falsifiable control-coordinate hypothesis, not as a universal ontology.

The article distinguishes exact results, conditional propositions, constructions, engineering hypotheses, and unresolved extensions. Among the latter are automatic boundary discovery, canonical factorization, composition of PORE cells, long-memory systems, Naming and Object formation, LuoShu-type trace structures, and prime-like decomposition under emergent composition grammars.

The central proposal is therefore not that one universal partition of reality exists, but that bounded observers can construct, test, revise, and translate operational worlds through a recursive interaction of boundaries, quotients, effective dynamics, intervention, and residual.




## Keywords

World Formation; framing; bounded observer; quotient dynamics; behavioral equivalence; coarse-graining; operational worlds; PORE; semantic field theory; intervention; residual; reframing; multiple frames; system identification.

# 1. Introduction

Scientific and practical reasoning normally begins after a large number of hidden decisions have already been made.

A physicist speaks of a “system” and its “environment.” An economist selects a market, a set of firms, and a collection of macroeconomic variables. A physician treats a patient as the principal unit while regarding temperature, diet, medication, and pathogens as external influences. An artificial agent receives a representation containing variables, objects, actions, and goals.

Yet none of these decompositions is logically prior to observation.

Before prediction can occur, an observer must already have decided, explicitly or implicitly,

- what lies inside the current domain of analysis;
- which differences between underlying states should be retained;
- which differences may be ignored;
- which operations count as admissible interventions;
- which outputs matter to the current task;
- and which failures should trigger parameter repair rather than a revision of the frame itself.

The problem is therefore deeper than ordinary model fitting.

A model estimates relationships within a given representation. Framing determines what the representation is allowed to distinguish in the first place.

This article develops an operational account of that prior layer.

The central claim is deliberately modest:

> A frame need not be understood as an ontology. It can first be treated as an operational construction specifying a boundary and a quotient of the states within that boundary.

The minimal framing kernel will therefore be written as

$$\Gamma_{min} = (B, \Pi). \quad (1.1)$$

Here $B$ is an operational boundary, while $\Pi$ is a projection or quotient map that determines which distinctions remain visible.

The corresponding effective state space is

$$\Pi : X_B \to Z. \quad (1.2)$$

Two microscopic states $x$ and $y$ are operationally identified whenever

$$\Pi(x) = \Pi(y). \quad (1.3)$$

Equivalently,

$$x \sim_\Pi y \Leftrightarrow \Pi(x) = \Pi(y). \quad (1.4)$$

Hence the effective state space may be viewed as a quotient,

$$Z \cong X_B / \sim_\Pi. \quad (1.5)$$

This formulation immediately separates framing from ontology.

Nothing in Equations (1.1)–(1.5) requires the equivalence classes of Z to be declared “objects,” assigned names, or interpreted as metaphysically fundamental entities. The quotient merely records which distinctions the present protocol continues to treat as operationally relevant.

This distinction will be essential throughout the article.

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# 1.1 The problem is not to discover one final partition

A tempting formulation of the framing problem is:

> What is the correct decomposition of reality?

The present framework does not begin with that assumption.

Different observers may use different partitions of the same underlying dynamics. Different scientific traditions may retain different distinctions. Different cultures may carry inherited interpretive decompositions into new environments. A decomposition formed in one context may remain surprisingly effective in another, even if it is not the locally simplest representation.

The relevant question is therefore not necessarily whether a partition is unique.

A more operational question is:

> Does the frame preserve the distinctions required for successful prediction, intervention, admissibility judgment, and measurement within the current task?

If it does, then it may remain a legitimate frame even when another observer uses a different decomposition.

This motivates a theory not of one mandatory World, but of operational frames, translations between frames, and the conditions under which a frame must be revised.

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# 1.2 Five levels that must not be conflated

Much confusion arises because several logically different operations are all described using words such as “boundary,” “category,” or “distinction.”

The present framework separates at least five levels:

1. World Formation or admission: which states satisfy a declared condition?
2. Operational cut: what is treated as inside and outside?
3. Quotient: which differences inside the boundary continue to count as differences?
4. Dynamics: does the effective state evolve autonomously enough to support prediction and control?
5. Reframing: what happens when the retained distinctions cease to be sufficient?

These operations interact, but they are not identical.

In particular,

$$\text{Cut} \neq \text{Quotient} \neq \text{Admission}. \quad (1.6)$$

The rest of this article develops the consequences of this separation.

# 2. Absolute World Formation: The Minimal Boundary Kernel

The most primitive level considered here is not yet a frame.

Let X be a domain of possible states and let $\Lambda$ index a family of admissibility declarations. For each $\lambda \in \Lambda$, define

$A_\lambda : X \to \{0, 1\}$. (2.1)

The admitted region is

$\Omega_\lambda = \{x \in X : A_\lambda(x) = 1\}$. (2.2)

A nontrivial World Formation event requires at least one $\lambda$ such that

$\emptyset \neq \Omega_\lambda \neq X$. (2.3)

The corresponding minimal kernel may therefore be written

$K_{abs} = (X, \Lambda, A)$. (2.4)

This construction is intentionally weak.

It establishes only that a nontrivial distinction between admitted and non-admitted states can be made.

It does not yet specify:

- a physical boundary;
- an environmental interface;
- a coarse-graining;
- a dynamical model;
- an observer identity;
- an object ontology;
- a PORE representation;
- a temporal arrow;
- or a unique decomposition of the state space.

This weakness is useful.

It prevents later structure from being silently inserted into the foundational definition.

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# 2.1 Admission is not the same as inside–outside

Suppose a state x lies within the system currently being modeled, but violates a viability condition.

Then one may simultaneously have

$x \in X_B$ (2.5)

and

$A(x) = 0$. (2.6)

The state is inside the modeled domain but inadmissible.

Therefore,

outside the operational boundary ≠ inadmissible state. (2.7)

This distinction becomes important in biological, organizational, financial, and AI systems, where failed, unstable, prohibited, or unsafe states still belong to the system's effective state space.

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## 2.2 Static admission alone does not generate history

If two static admission operators act by intersection,

$B_A(S) = S \cap \Omega_A$, (2.8)

$B_B(S) = S \cap \Omega_B$, (2.9)

then

$B_A(B_B(S)) = B_B(B_A(S)) = S \cap \Omega_A \cap \Omega_B$. (2.10)

Thus purely static filters commute.

They do not by themselves generate order dependence, history, or temporal direction.

A time-bearing theory therefore requires additional structure: changing admissibility conditions, state-changing disclosure events, ledgers, traces, or other forms of nontrivial sequential dependence.

This earlier result remains compatible with the framing theory developed below: static World Formation supplies a distinction, while dynamic framing determines how retained distinctions evolve.

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# 3. The Minimal Framing Kernel

A World Formation distinction does not yet determine how the admitted or modeled region should be represented.

For that, two additional choices are required.

First, one needs an operational boundary B. This determines what is currently treated as internal to the analysis and what is treated as environmental.

Second, one needs a projection Π. This determines which distinctions within the internal domain are retained.

The minimal frame is therefore

$\Gamma_{min} = (B, \Pi)$. (3.1)

Let $X_B$ denote the microscopic state space inside B.

Then

$\Pi : X_B \to Z.$ (3.2)

The effective state $z \in Z$ is

$z = \Pi(x).$ (3.3)

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# 3.1 Framing as quotienting

Define

$x \sim_\Pi y \Leftrightarrow \Pi(x) = \Pi(y).$ (3.4)

The relation $\sim_\Pi$ partitions $X_B$ into equivalence classes.

The effective state space is therefore

$Z \cong X_B / \sim_\Pi.$ (3.5)

This formulation produces an important change of perspective.

A frame is often described by asking what information it preserves.

Equally important is what the frame declares irrelevant.

Every equivalence class $[x]_\Pi$ contains microscopic states that the current frame no longer distinguishes.

Thus the projection encodes an operational statement:

> Differences within one fiber of $\Pi$ do not currently matter enough to justify separate effective states.

This does not mean that those microscopic differences do not exist.

It means only that the current frame treats them as equivalent.

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# 3.2 Framing is therefore partly an act of forgetting

Suppose $X_B$ is locally $n$-dimensional while the effective representation $Z$ is $d$-dimensional, with

$\operatorname{rank}(D\Pi) = d < n.$ (3.6)

Then at least $n - d$ local directions have been removed from the effective description.

A crude dimensional compression ratio would be

$c_\Pi = 1 - d/n.$ (3.7)

This measure is generally inadequate for nonlinear, stochastic, discrete, or information-theoretic systems, but it illustrates the main point:

> Coarse-graining reduces the number of distinctions treated as operationally independent.

More sophisticated measures may use entropy, description length, predictive information, control complexity, or task-relative distinguishability.

No specific compression measure is assumed at the Core level.

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# 3.3 Compression alone is not a valid frame

Consider the trivial projection

$\Pi(x) = z_0$ for every $x \in X_B$. (3.8)

This achieves maximal compression.

It also destroys nearly all predictive and control information.

Thus:

Compression does not imply closure. (3.9)

A useful projection must satisfy additional compatibility conditions with the dynamics, interventions, admissibility criteria, and measurements relevant to the task.

These conditions will now be made explicit.

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# 4. Behavioral Equivalence and the Coarsest Exact Operational Frame

Consider a deterministic controlled system on the fixed operational domain $X_B$:

$x_{t+1} = F_u(x_t), u \in U$. (4.1)

Let

$o : X_B \to Y$ (4.2)

be the task-relevant observation map.

The map $o$ may include any distinctions that must be retained by the protocol, such as measurements, task outputs, or admissibility indicators.

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# 4.1 Intervention words

Let

$w = (u_1, u_2, \ldots, u_n)$ (4.3)

be a finite sequence of admissible interventions.

Define the corresponding composed dynamics

$F_w = F_{u_n} \circ \ldots \circ F_{u_2} \circ F_{u_1}$. (4.4)

The empty word $\varepsilon$ is defined by

$F_\varepsilon = \mathrm{id}$. (4.5)

This allows present and future task-relevant behavior to be tested within one definition.

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# 4.2 Exact behavioral equivalence

Two states x and y are behaviorally equivalent under protocol P if no admissible finite intervention sequence can distinguish them through the observation map.

Define

$x \equiv_P y \Leftrightarrow o(F_w(x)) = o(F_w(y))$ for every finite admissible word w. (4.6)

This relation is reflexive, symmetric, and transitive, and therefore defines an equivalence relation on $X_B$.

The corresponding quotient is

$Z_P^* = X_B / \equiv_P$. (4.7)

The canonical projection is

$\Pi_P^* : X_B \to Z_P^*$. (4.8)

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# Theorem 1 — Coarsest Exact Operational Quotient

For fixed $X_B$, U, F, and o, the quotient $Z_P^*$ defined by Equation (4.7) is the coarsest exact quotient that preserves all task-relevant behavior under all finite admissible intervention sequences.

## Proof

Let $\Pi : X_B \to Z$ be any exact operational projection that preserves o and admits well-defined effective dynamics for every $u \in U$.

Suppose

$\Pi(x) = \Pi(y)$. (4.9)

Because the effective dynamics are well-defined on Z, equality of the projected states implies equality after any finite admissible intervention sequence:

$\Pi(F_w(x)) = \Pi(F_w(y))$. (4.10)

Because the observation map is preserved by the projection, there exists an effective observation map $\bar{o}$ such that

$o = \bar{o} \circ \Pi$. (4.11)

Therefore,

$o(F_w(x)) = o(F_w(y))$ (4.12)

for every finite admissible word w.

Hence,

$x \equiv_P y$. (4.13)

Thus any exact operational quotient $\Pi$ can identify only states already identified by $\equiv_P$.

Therefore $Z_{P^*}$ is the coarsest exact operational quotient. ■

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# 4.3 What the theorem does and does not establish

The theorem is stronger than ordinary dimensionality reduction.

It does not say merely that certain variables are statistically redundant.

It says that two states may be identified only if no allowed future experimental sequence can distinguish them through the outputs relevant to the protocol.

The resulting quotient therefore has a precise operational meaning.

However, the theorem does not establish a universal ontology.

The quotient depends on the chosen protocol:

$P = (B, U, o)$. (4.14)

Change the boundary, intervention set, or observation map, and the equivalence relation may change.

Thus the theorem establishes uniqueness only relative to a fixed protocol.

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# 5. When Dynamics Descend to the Quotient

The previous section defined behavioral equivalence directly from all admissible future probes. A related result begins from an arbitrary candidate projection $\Pi$ and asks whether the microscopic dynamics induce a well-defined effective dynamics on its quotient.

Suppose

$\Pi : X_B \to Z$. (5.1)

We seek effective maps

$K_u : Z \to Z$ (5.2)

such that

$\Pi \circ F_u = K_u \circ \Pi$. (5.3)

This commuting condition is the mathematical expression of exact dynamical closure.

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# Theorem 2 — Exact Quotient Dynamics

If, for every $x, y \in X_B$ and every admissible $u$,

$\Pi(x) = \Pi(y) \Rightarrow \Pi(F_u(x)) = \Pi(F_u(y))$, (5.4)

then there exists a unique induced map $K_u$ on $Z$ satisfying

$\Pi \circ F_u = K_u \circ \Pi$. (5.5)

### Proof

For $z \in Z$, choose any representative $x$ satisfying

$\Pi(x) = z$. (5.6)

Define

$K_u(z) = \Pi(F_u(x))$. (5.7)

Condition (5.4) guarantees that this value does not depend on the chosen representative $x$.

Therefore $K_u$ is well-defined.

Uniqueness follows because every $z \in Z$ has at least one representative in $X_B$. ■

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# 5.1 Dynamic congruence

Condition (5.4) can be interpreted directly:

> Once two microscopic states have been declared equivalent, the dynamics must not immediately reveal a distinction that the quotient has erased.

A projection violating this condition may still provide useful compression, but it is not an exact closed dynamical frame.

This gives the first major framing residual:

$r_D$ = dynamical closure defect. (5.8)

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# 5.2 Intervention congruence

Natural evolution alone may not be sufficient.

Two microscopic states may behave similarly without intervention but respond differently when actively probed.

Therefore a control-capable frame must preserve intervention-relevant distinctions.

Abstractly, one may define

$r_I$ = intervention inconsistency defect. (5.9)

A frame with small $r_D$ but large $r_I$ may be adequate for passive prediction but inadequate for control.

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# 5.3 Admission congruence

Suppose microscopic admissibility is described by

$A_X : X_B \to \{0, 1\}$. (5.10)

If

$\Pi(x) = \Pi(y)$ (5.11)

but

$A_X(x) \neq A_X(y)$, (5.12)

then the effective state $\Pi(x)$ cannot tell whether it is admitted.

For admissibility to descend to the quotient, one requires

$\Pi(x) = \Pi(y) \Rightarrow A_X(x) = A_X(y)$. (5.13)

Under this condition there exists

$A_Z : Z \to \{0, 1\}$ (5.14)

such that

$A_X = A_Z \circ \Pi$. (5.15)

This gives a second form of compatibility between World Formation and framing.

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# 5.4 Measurement congruence

Suppose a task-relevant measurement is

$M : X_B \to Y_M$. (5.16)

For measurement to be representable at the effective level, one requires

$M = \bar{M} \circ \Pi$ (5.17)

for some effective measurement map $\bar{M}$.

Failure produces another residual,

$r_M$ = measurement sufficiency defect. (5.18)

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# 5.5 A typed projection residual

The projection should therefore not be judged by one undifferentiated error measure.

A useful residual vector is

$R_\Pi = (r_D, r_I, r_A, r_M)$. (5.19)

Here:

- $r_D$ measures dynamic closure failure;

- r_I measures intervention inconsistency;
- r_A measures admission inconsistency;
- r_M measures measurement insufficiency.

A frame is operationally admissible only relative to a tolerance profile ε:

$R_\Pi \preceq \varepsilon$. (5.20)

This typed residual will later become the main diagnostic for deciding whether a failure should be repaired within the current frame or should instead trigger reframing.

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# 6. The Coarsest Sufficient Frame as an Optimization Principle

If zero residual were the only objective, the identity projection

$\Pi = id_X$ (6.1)

would always be safe.

But it would perform no coarse-graining.

Framing therefore balances sufficiency against representational complexity.

A generic formulation is

minimize $C(\Pi)$ subject to $R_\Pi \preceq \varepsilon$. (6.2)

The desired projection is therefore not the most informative representation.

It is the coarsest representation that remains adequate for the task.

Symbolically,

$\Pi^*$ = coarsest task-sufficient quotient. (6.3)

This principle may be summarized as:

> Retain a distinction only when erasing it would push some task-relevant residual beyond tolerance.

This formulation will become especially important when bounded observers, approximate equivalence, and frame plurality are introduced in the next sections.

# 7. Bounded Observers and Provisional Frames

The exact behavioral quotient of Section 4 assumes an observer capable of evaluating every finite admissible intervention sequence. Real observers cannot do this.

A bounded observer has at least four restrictions:

- finite probing depth;
- finite intervention repertoire;
- finite measurement resolution;
- finite computational and experimental budget.

Let H denote the maximum intervention depth currently available to the observer. Define finite-depth behavioral equivalence by

$$x \equiv_H y \Leftrightarrow o(F_w(x)) = o(F_w(y)) \text{ for every admissible } w \text{ with } |w| \le H. \quad (7.1)$$

This generates a sequence of equivalence relations,

$$\equiv_0 \supseteq \equiv_1 \supseteq \equiv_2 \supseteq \dots \supseteq \equiv_\infty. \quad (7.2)$$

The corresponding effective spaces are

$$Z_H = X_B / \equiv_H. \quad (7.3)$$

As the observer gains access to deeper probes, previously merged states may become distinguishable.

Thus bounded observation naturally creates a filtration of frames.

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# 7.1 Behavioral framing filtration

At depth zero, two states are equivalent whenever their currently visible outputs agree.

At depth one, they must also remain indistinguishable after every single admissible intervention.

At depth two, every two-step intervention sequence must fail to distinguish them.

The sequence therefore refines monotonically:

$$\equiv_0 \supseteq \equiv_1 \supseteq \equiv_2 \supseteq \dots . \quad (7.4)$$

Correspondingly,

$$Z_0 \leftarrow Z_1 \leftarrow Z_2 \leftarrow \dots . \quad (7.5)$$

The effective world becomes more finely partitioned as experimental depth increases.

This motivates an important interpretation:

> Boundedness creates provisional equivalences.

A pair of states may be equivalent for the current observer without being behaviorally identical in the exact limit.

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# 7.2 Learning as partition refinement

Suppose the current observer treats

$$x \equiv_H y. \quad (7.6)$$

A newly available intervention sequence $w^*$ reveals

$$o(F_{w^*}(x)) \neq o(F_{w^*}(y)). \quad (7.7)$$

The previous equivalence class must then be split.

Symbolically,

[x]$H \to$ *[x]*{H+1}^{(1)} ∪ [x]_{H+1}^{(2)}. (7.8)

This gives a precise representation of one important form of learning:

> Learning is the refinement of operational equivalence classes in response to newly discriminating evidence.

This is stronger than merely updating parameters inside a fixed model.

The representational state space itself has changed.

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# 7.3 Recursive depth is not physical time

The index H measures discrimination depth, not physical time.

Although an intervention sequence unfolds temporally, the hierarchy

H = 0, 1, 2, … (7.9)

describes how deeply the observer is willing or able to probe consequences.

It must therefore be distinguished from both physical time and recursive model depth:

physical time ≠ probing depth ≠ revision depth. (7.10)

This separation will matter later when trace, memory, and self-revision are considered.

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# 8. Approximation and the Emergence of Frame Plurality

Exact equivalence is mathematically convenient but empirically restrictive.

Real systems contain noise, finite precision, stochasticity, and approximate rather than exact closures.

A natural approximate condition is

d_Y(o(F_w(x)), o(F_w(y))) ≤ ε (8.1)

for all admissible probe sequences within some specified class.

One might then define

x ~_ε y (8.2)

whenever their behavioral distance lies below tolerance.

However, a major difficulty immediately appears.

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# No-Go 1 — Approximate indistinguishability need not be transitive

It may happen that

$d_B(x, y) < \varepsilon$, (8.3)

$d_B(y, z) < \varepsilon$, (8.4)

but

$d_B(x, z) > \varepsilon$. (8.5)

Therefore,

$x \sim_\varepsilon y$ and $y \sim_\varepsilon z$ (8.6)

does not necessarily imply

$x \sim_\varepsilon z$. (8.7)

Hence tolerance-based similarity need not define an equivalence relation.

A quotient cannot be formed directly without an additional partition policy.

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# 8.1 Why bounded approximation creates multiple legitimate frames

Consider three states x, y, and z satisfying Equations (8.3)–(8.5).

Two admissible partitions might be

$P_A = \{\{x, y\}, \{z\}\}$, (8.8)

and

$P_B = \{\{x\}, \{y, z\}\}$. (8.9)

Neither partition is forced uniquely by the approximate similarity relation.

If both satisfy the required tolerances for the task, then both may be operationally admissible.

Frame plurality therefore need not arise from philosophical relativism.

It can arise directly from approximation.

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# 8.2 Exact uniqueness versus bounded plurality

The distinction can now be stated sharply.

For a fixed exact protocol,

$P = (B, U, o)$, (8.10)

there exists a unique coarsest exact behavioral quotient.

For a bounded approximate observer,

$P_O = (B, U_O, o_O, H_O, \varepsilon_O)$, (8.11)

multiple incomparable admissible frames may exist.

Thus,

exact operational framing may be unique, while bounded operational framing may be plural. (8.12)

This distinction is central to the rest of the framework.

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# 9. Plural Frames Without Relativism

Frame plurality does not imply that all frames are equally successful.

Different frames can be judged by prediction, intervention, admissibility preservation, empirical fit, complexity, robustness, and translation cost.

Suppose two observers construct

$\Pi_A : X \to Z_A$, (9.1)

and

$\Pi_B : X \to Z_B$. (9.2)

There is no requirement that

$Z_A = Z_B$. (9.3)

The relevant question is whether their operational structures can be translated.

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# 9.1 Cross-frame translation

Let

$T_{AB} : Z_A \to Z_B$ (9.4)

be a translation map.

A strong form of dynamical compatibility is

$T_{AB} \circ K_A \approx K_B \circ T_{AB}$. (9.5)

Likewise, one may require approximate preservation of admissibility and measurement:

$A_B(T_{AB}(z)) \approx A_A(z)$, (9.6)

$M_B(T_{AB}(z)) \approx T_M(M_A(z))$. (9.7)

If these relations hold within the task tolerance, the two frames may differ substantially while remaining operationally compatible.

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# 9.2 A frame can travel

This has an important consequence.

A frame formed in one environment may remain useful in another.

The observer need not abandon the original decomposition merely because another culture, discipline, or engineering tradition uses a different one.

What matters is whether the imported frame can still:

- predict effectively;
- support intervention;
- translate into locally relevant distinctions;
- and keep its residuals within acceptable limits.

Thus one may retain a prior perspective while learning a new operational world.

This motivates an atlas rather than a single mandatory decomposition.

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## 9.3 Common quotients and partial agreement

Two observers need not agree at their finest level.

Suppose there exists a third representation Z_C and maps

g_A : Z_A → Z_C, (9.8)

g_B : Z_B → Z_C (9.9)

such that

g_A ∘ Π_A = g_B ∘ Π_B. (9.10)

Then the two observers possess a common coarse quotient.

Agreement therefore need not mean identical framing.

It may mean only that different frames share a stable common projection.

This provides one operational route toward inter-observer objectivity without demanding a unique universal frame.

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# 10. Boundary Placement as a Higher-Level Problem

The exact quotient theorem assumed that B had already been fixed.

In practice, the location of the operational boundary may itself be uncertain.

A degree of freedom q can sometimes be represented either as internal state or as an environmental input.

For one boundary,

q ∈ X_B. (10.1)

For another,

$q \in E_B$. (10.2)

Boundary movement therefore changes the type of a variable:

internal state ↔ environmental input. (10.3)

This means that boundary choice and quotient choice are not independent.

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# 10.1 State–interface trade-off

A useful engineering construction is to evaluate a candidate boundary B by three broad costs:

$C_\Gamma(B) = C_{state}(B) + C_{interface}(B) + C_{residual}(B)$. (10.4)

Here:

- $C_{state}$ measures the complexity of the effective internal representation;
- $C_{interface}$ measures the complexity of the information crossing the boundary;
- $C_{residual}$ measures closure failure and unmodeled interactions.

Moving the boundary outward usually increases internal state burden but may simplify the interface.

Moving it inward may reduce internal state burden while forcing more environmental information through the interface.

This creates a state–interface trade-off.

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# 10.2 Boundary-knee hypothesis

Consider a nested family of candidate boundaries

$B_1 \subset B_2 \subset \ldots \subset B_n$. (10.5)

A local boundary knee $B_{k^*}$ satisfies

$C_\Gamma(B_{k^*}) < C_\Gamma(B_{k^*-1})$ (10.6)

and

$C_\Gamma(B_{k^*}) < C_\Gamma(B_{k^*+1})$. (10.7)

The interpretation is simple:

> At the boundary knee, enlarging the inside increases internal complexity more than it reduces interface burden, while shrinking the inside increases interface burden more than it saves internal complexity.

This is not claimed as a universal theorem.

It is an engineering hypothesis for identifying useful operational boundaries.

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# 10.3 Boundary existence, adequacy, and identifiability

Three levels should be distinguished:

Boundary existence < Boundary adequacy < Boundary identifiability. (10.8)

A boundary exists whenever an inside–outside cut is specified.

It is adequate when the resulting frame performs sufficiently well.

It is identifiable only if available evidence distinguishes that cut from relevant alternatives.

A useful effective model therefore does not automatically imply that its physical boundary is uniquely determined.

---

# No-Go 2 — Minimal realization does not imply unique physical boundary

Two different boundaries may generate the same external input–output behavior.

Thus,

same effective transducer ⇏ same microscopic cut. (10.9)

A minimal operational realization may be identifiable up to coordinate transformation even when its embedding into the underlying physical variables is not.

This distinction prevents a successful model from being mistaken for a proof of a unique ontology.

---

# 11. Repair and Reframing

A central practical question is whether a model failure should be repaired inside the current frame or whether the frame itself should change.

Let a working model be

$\Gamma = (B, \Pi, \theta)$, (11.1)

where $\theta$ denotes parameters of the effective dynamics.

---

## 11.1 Repair

If the projection and boundary remain valid but parameter values are incorrect, one may update

$\theta \to \theta'$. (11.2)

This is repair.

The effective distinctions remain unchanged.

---

## 11.2 Quotient reframing

Suppose instead that two states satisfy

$\Pi(x) = \Pi(y)$, (11.3)

but a new admissible probe shows

$o(F_w(x)) \neq o(F_w(y))$. (11.4)

The problem cannot be repaired by merely changing $\theta$.

The equivalence relation itself is too coarse.

One must refine the projection:

$\Pi \to \Pi'$. (11.5)

This is quotient reframing.

---

## 11.3 Boundary reframing

A different failure occurs when an important variable has been placed on the wrong side of the boundary.

Then the correction is

$B \to B'$. (11.6)

The variable may move from environment to internal state, or vice versa.

This is boundary reframing.

---

## 11.4 Deep reframing

The general update is therefore

$(B, \Pi, \theta) \to (B', \Pi', \theta')$. (11.7)

When both boundary and quotient change substantially, the revision is deeper than ordinary parameter learning.

This provides a formal basis for distinguishing:

- repair;
- refinement;
- boundary migration;
- and deep reframing.

---

## 11.5 Residual must be typed

A scalar residual is not sufficient to decide which kind of revision is required.

A more informative framing residual is

R_F = (r_B, r_D, r_I, r_A, r_M, r_H). (11.8)

Possible components include:

- r_B: boundary defect;
- r_D: dynamic closure defect;
- r_I: intervention defect;
- r_A: admissibility defect;
- r_M: measurement defect;
- r_H: memory or hidden-history defect.

Different residual signatures imply different repairs.

For example:

high r_D with low r_B may suggest projection refinement; (11.9)

high r_H may suggest memory augmentation rather than new instantaneous coordinates; (11.10)

high r_B may indicate that the boundary itself should move. (11.11)

Thus residual is not merely error.

It is diagnostic pressure on the frame.

---

# 12. The Self-Revising Framing Kernel

The preceding results can now be assembled into a compact recursive architecture.

Start with

Γ_n = (B_n, Π_n). (12.1)

Infer or estimate effective dynamics

K_n. (12.2)

Evaluate the typed residual

R_n. (12.3)

Use the residual to generate a revised frame,

Γ_{n+1} = F_frame(Γ_n, R_n, E_n), (12.4)

where E_n denotes newly acquired evidence.

The core loop is therefore

(B, Π) → K → R → (B', Π'). (12.5)

This will be called the Self-Revising Framing Kernel.

---

# 12.1 Generate, regulate, admit, declare

The same loop can be written at the level of candidate frames.

Let G_F generate candidate framings:

$G_F(X, T) = \{\Gamma_1, \Gamma_2, \ldots\}$. (12.6)

Let R_F evaluate their typed residuals.

A frame is admitted when

$A_F(\Gamma) = 1[R_F(\Gamma) \preceq \varepsilon_F]$. (12.7)

The admissible set is

$\Omega_F = \{\Gamma : A_F(\Gamma) = 1\}$. (12.8)

If operational circumstances require commitment to one working frame, a declaration rule may select

$D_F : \Omega_F \to \Gamma^*$. (12.9)

This produces the architecture

Generate → Regulate → Admit → Declare. (12.10)

Importantly, Declaration is not always required.

If several frames remain admissible and no action requires immediate commitment, the system may preserve a frame atlas rather than forcing premature uniqueness.

---

## 12.2 Framing itself as a lifted World Formation process

At the lowest level, World Formation applies admissibility to states.

At the framing level, the same logical shape can be applied to candidate representations.

The domain is now a space of frames,

$\mathfrak{F} = \{\Gamma\}$. (12.11)

An admissibility operator selects

$\Omega_F \subset \mathfrak{F}$. (12.12)

Thus framing may itself be regarded as World Formation lifted from state space to frame space.

This does not mean that all framing problems have been solved by the original admission operator.

It means that the same general architecture of generation, constraint, admission, and revision can recur at a higher type level.

---

# 13. A Single-Structure Specialization

The general framing kernel does not require any particular choice of coordinates.

However, many practical problems arrive with an approximate operational boundary already given.

Examples include:

- a firm;
- a patient;
- a machine;
- a market;
- a project;
- an agent;
- a bounded ecological unit.

In such cases, one can search for a more specialized low-dimensional framing.

A useful candidate takes the form

$\Gamma_{SS} = (B, \Delta_E, \eta, \Xi, K)$. (13.1)

Here:

- $B$ is the given or provisionally accepted boundary;
- $\Delta_E$ is the principal environmental drive;
- $\eta$ collects secondary environmental modes;
- $\Xi$ is the internal coarse state;
- $K$ is the effective dynamics.

This is not the general definition of a frame.

It is a structured specialization for bounded single-structure problems.

---

# 13.1 Environmental drive extraction

Let $e \in \mathbb{R}^m$ represent environmental variables and $p$ the internal effective state.

Suppose

$\dot{p} = F(p, e)$. (13.2)

The local sensitivity to environmental forcing is

$J_E = \partial F / \partial e$. (13.3)

Take a singular value decomposition,

$J_E = U \Sigma V^T$. (13.4)

If

$\sigma_1 \gg \sigma_2$, (13.5)

then the environmental influence is approximately rank one.

The principal drive may be defined as

$\Delta_E = v_1^T e$. (13.6)

The leading environmental effect is then

$J_E e \approx \sigma_1 u_1 \Delta_E$. (13.7)

This gives an operational interpretation of a Heaven–Earth axis:

> Heaven–Earth is a boundary-relative dominant environmental polarity, not two independent dimensions.

Two reference anchors may be chosen along $\Delta_E$,

$E^+$ and $E^-$, (13.8)

representing opposite ends of the dominant environmental drive.

---

# 13.2 Residual environmental modes

A one-axis environmental description is only justified if the remaining singular directions are sufficiently weak.

Define

$\eta_k = v_k^T e$ for $k \geq 2$. (13.9)

An effective environmental rank may be defined by

$r_E(\varepsilon) = \min\{r : \Sigma_{k>r} \sigma_k^2 / \Sigma_k \sigma_k^2 \leq \varepsilon\}$. (13.10)

The single-axis specialization is appropriate only when

$r_E(\varepsilon) = 1$. (13.11)

If not, one should retain multiple environmental drives rather than forcing all environmental influence into one scalar.

---

# Counterexample — Rank-two forcing

Consider

$\dot{p}_1 = -p_1 + e_1$, (13.12)

$\dot{p}_2 = -p_2 + e_2$, (13.13)

$\dot{p}_3 = -p_3$. (13.14)

The environmental sensitivity has rank two.

No scalar function $\Delta_E$ can preserve both independent forcing directions for tasks requiring separate control of $p_1$ and $p_2$.

Thus:

one Heaven–Earth axis is a testable low-rank hypothesis, not a universal requirement. (13.15)

---

# 14. PORE as a Candidate Internal Quotient

For the internal state of a bounded single-structure system, consider the coarse projection

$\Pi_{PORE} : X_I \to \Xi$, (14.1)

with

$\Xi = (\rho, \gamma, \tau)$. (14.2)

The three coordinates are defined by functional roles rather than by microscopic ontology.

$\rho$ represents effective occupancy, content, density, or amount retained within the operational structure.

$\gamma$ represents constraint, confinement, domain-locking, or effective holding strength.

$\tau$ represents agitation, dephasing, fluctuation, perturbation, or coherence loss.

The claim is not that physical reality has three dimensions of this form.

The hypothesis is instead that many bounded open systems may admit a low-dimensional regime or control description in which these three operational roles are sufficient.

---

# 14.1 PORE is not full-state reconstruction

The map

$\Pi_{PORE} : X_I \to \mathbb{R}^3$ (14.3)

is intended as a coarse control or regime representation.

It need not reconstruct:

- spatial heterogeneity;
- vector flows;
- anisotropic stress;
- detailed microscopic distributions;
- long history;
- or all latent variables.

A richer field description may still contain scalar, vector, and tensor quantities that are subsequently coarse-grained into $\Xi$.

Thus the strongest defensible PORE claim is currently one of control or regime sufficiency, not ontological completeness.

---

# 14.2 Why three?

The number three should not be justified merely by repeated empirical convenience.

A stronger question is whether a class of tasks requires three locally independent coarse roles.

Consider the noisy confinement model

$$dX_t = -\gamma X_t\,dt + \sqrt{(2\tau)}\,dW_t. \quad (14.4)$$

Let $\rho$ represent an independent effective occupancy coordinate.

Consider the task outputs

$$Y(\rho, \gamma, \tau) = (\rho, 1/\gamma, \tau/\gamma). \quad (14.5)$$

The Jacobian is

$$DY = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\gamma^{-2} & 0 \\ 0 & -\tau\gamma^{-2} & \gamma^{-1} \end{pmatrix}. \quad (14.6)$$

Its determinant is

$$\det(DY) = -\gamma^{-3}. \quad (14.7)$$

For $\gamma \neq 0$,

$$\det(DY) \neq 0. \quad (14.8)$$

Therefore the task map has local rank three.

---

# Conditional Proposition 1 — A Three-Dimensional Minimality Witness

If an exact smooth representation $q \in \mathbb{R}^d$ preserves all three task outputs in Equation (14.5), then locally

$$\operatorname{rank}(DY) \leq d. \quad (14.9)$$

Since

$$\operatorname{rank}(DY) = 3, \quad (14.10)$$

one must have

$$d \geq 3. \quad (14.11)$$

Thus $\Xi = (\rho, \gamma, \tau)$ attains the minimum dimension for this specific class and task.

This does not prove that all systems require three variables.

It establishes only that three-dimensional minimality can arise naturally and non-arbitrarily.

---

# 14.3 Approximate minimality

For approximate tasks, let

$$J_\Xi = \partial Y/\partial \Xi. \quad (14.12)$$

Let its singular values be

$s_1 \geq s_2 \geq s_3$. (14.13)

If

$s_3 \gg \varepsilon_{task}$, (14.14)

then all three directions remain task-relevant.

If

$s_3 \approx 0$, (14.15)

the effective control dimension may collapse to two.

Thus:

three-dimensional necessity is regime-relative and task-relative. (14.16)

This preserves falsifiability.

---

# 15. Composition and the Re-Emergence of the PORE Triple

A central question is whether low-dimensional PORE descriptions survive composition.

Suppose two subsystems possess internal triples

$\Xi_A = (\rho_A, \gamma_A, \tau_A)$, (15.1)

$\Xi_B = (\rho_B, \gamma_B, \tau_B)$. (15.2)

Their microscopic composite is generally higher-dimensional.

Nothing guarantees that

$\Xi_A \times \Xi_B$ (15.3)

can be reduced again to a single triple.

Thus:

composition does not automatically preserve PORE form. (15.4)

A separate closure argument is required.

---

# 15.1 Five conditions for approximate recursive closure

A useful conditional construction requires at least the following.

### BR1 — Occupancy aggregability

There must exist an effective aggregate

$\rho^* = A_\rho(\rho_1, \ldots, \rho_N, q_{int})$, (15.5)

where any interface storage q_int is either fast or explicitly included.

### BR2 — Constraint homogenizability

Task-relevant retention and escape effects must admit a scalar effective description,

$\gamma^* = A_\gamma(\{\gamma_i\}, \{\lambda_{ij}\}, B^*)$. (15.6)

Persistent task-relevant anisotropy violates this condition.

### BR3 — Agitation scalarizability

The effective agitation must include relevant covariance and interface fluctuation terms,

$\tau^* = A_\tau(\{\tau_i\}, \mathrm{Cov}_{ij}, \text{interface fluctuations})$. (15.7)

### BR4 — Fast interface or finite memory

Unretained interface states must relax sufficiently quickly relative to the macro timescale, or their memory kernels must decay enough for a Markovian approximation.

### BR5 — Operational sufficiency

If two composite microscopic states produce the same $\Xi^*$, their task-relevant futures under admissible inputs must remain approximately indistinguishable.

---

## Conditional Proposition 2 — Boundary Recursion Closure

If BR1–BR5 hold for a composite system, then there exists an approximate macro compiler

$C^* : X_M \to \mathbb{R}^3$, (15.8)

with

$C^*(x) = \Xi^* = (\rho^*, \gamma^*, \tau^*)$, (15.9)

such that

$\Xi^*_{t+\Delta} = F^*(\Xi^*_t, u_t, \Delta_{E^*}) + \varepsilon^*$, (15.10)

with

$\|\varepsilon^*\| \le \varepsilon_{cl}$. (15.11)

This proposition is conditional.

Its importance lies not in proving that BR1–BR5 are universal, but in showing exactly what must be tested for a recursive PORE description to be valid.

---

## 15.2 PORE as a renormalization-stable family

If repeated composition and coarse-graining return the system approximately to the same family

of triples, then one may write

$$R(P_\Xi) \subseteq_\varepsilon P_\Xi. \quad (15.12)$$

This suggests a stronger but still empirical interpretation:

> PORE may define a renormalization-stable family of bounded control descriptions.

This is a more defensible claim than saying that all microscopic systems are fundamentally three-dimensional in PORE variables.

---

# 16. Minimal Counterexamples to PORE Closure

A useful theory should explain not only when its preferred representation works, but how it fails.

Several minimal counterexamples reveal distinct failure modes.

---

# 16.1 Two independent stocks

Let

$$\dot{a} = -\alpha a, \quad (16.1)$$

$$\dot{b} = -\beta b, \quad (16.2)$$

with

$$\alpha \neq \beta. \quad (16.3)$$

Suppose one defines

$$\rho = a + b. \quad (16.4)$$

Consider

$$x = (a, b) = (1, 0), \quad (16.5)$$

and

$$x' = (a, b) = (0, 1). \quad (16.6)$$

Both satisfy

$$\rho(x) = \rho(x') = 1. \quad (16.7)$$

But

$$\dot{\rho}(x) = -\alpha, \quad (16.8)$$

while

$$\dot{\rho}(x') = -\beta. \quad (16.9)$$

Therefore one scalar occupancy variable is not Markov sufficient.

The failure may be repaired by:

- retaining two stocks;
- adding a difference coordinate;
- or decomposing the system into coupled subsystems.

Which repair is preferable depends on the operational structure.

---

# 16.2 Persistent anisotropy

Consider

$$dx = -\gamma_x x\,dt + \sqrt{(2\tau)}\,dW_x, \quad (16.10)$$

$$dy = -\gamma_y y\,dt + \sqrt{(2\tau)}\,dW_y. \quad (16.11)$$

A scalar summary such as

$$\bar{\gamma} = (\gamma_x + \gamma_y)/2 \quad (16.12)$$

cannot preserve directional recovery when $\gamma_x$ and $\gamma_y$ matter independently.

For example,

$$(\gamma_x, \gamma_y) = (1, 9) \quad (16.13)$$

and

$$(\gamma_x, \gamma_y) = (9, 1) \quad (16.14)$$

have the same mean but respond differently to directional interventions.

The natural repair may be

$$\gamma \to (\gamma_x, \gamma_y), \quad (16.15)$$

or a tensorial confinement variable.

Thus a failure of scalar $\gamma$ does not automatically imply a new peer-level primitive of the same kind.

---

# 16.3 Colored noise and hidden memory

Consider

$$\dot{x} = -\gamma x + \eta, \quad (16.16)$$

and

$$d\eta = -\eta/\theta\,dt + \sigma\,dW_t. \quad (16.17)$$

Two states may share the same $x$ and the same coarse agitation magnitude $\tau$ while having different current $\eta$.

Their immediate futures then differ.

If

$\theta \ll t_{macro}$, (16.18)

$\eta$ may be eliminated approximately, producing effective white noise.

If

$\theta \sim t_{macro}$ (16.19)

or larger, $\eta$ becomes a relevant slow state or memory variable.

The failure is therefore not necessarily evidence for a fourth instantaneous PORE coordinate.

It may indicate that the state must be lifted to include history.

---

## 16.4 Rank-two environmental forcing

The rank-two example of Section 13 shows another type of failure.

If

$\mathrm{rank}(J_E) = 2$, (16.20)

one principal environmental scalar cannot preserve all task-relevant forcing.

The proper repair may be

$\Delta_E \to (\Delta_1, \Delta_2)$, (16.21)

or a decomposition into multiple driven subsystems.

This is an environmental failure, not necessarily an internal PORE failure.

---

## 16.5 Type before dimension

These counterexamples support a general diagnostic rule:

$d_{eff} > 3 \not\Rightarrow$ a fourth PORE primitive. (16.22)

The missing degree of freedom may instead be:

- an additional stock;
- an anisotropic component;
- a memory variable;
- a second environmental drive;
- a hidden subsystem;
- or evidence that the boundary itself is misplaced.

Hence:

> Type before dimension.

This rule prevents every closure failure from being converted into uncontrolled primitive inflation.

# 17. Factorization of the Effective State Space

The quotient construction determines which microscopic distinctions survive into the effective state space,

$Z = X_B / \sim$. (17.1)

It does not yet determine whether Z itself has a preferred decomposition into coordinates or subsystems.

A further question is therefore:

> When does the effective state space admit a meaningful factorization?

Formally, one may ask whether

$Z \cong Z_1 \times Z_2 \times \dots \times Z_k$. (17.2)

This question must be separated from dimensionality alone.

If

$\dim Z = k$, (17.3)

it does not follow that Z possesses k privileged factors.

An invertible transformation

$z' = Tz$ (17.4)

may mix all coordinates while preserving the dimension of the effective state space.

Thus:

$\dim Z = k \not\Rightarrow$ a privileged k-factor decomposition. (17.5)

The problem is not how many coordinates exist, but whether different coordinates possess distinct operational roles that remain stable under changes of representation.

---

# 17.1 Set factorization is weaker than operational factorization

Suppose a finite effective state space has nine states.

Since

$9 = 3 \times 3$, (17.6)

one can always assign the nine states to positions of a $3 \times 3$ array.

This establishes only a set-theoretic bijection.

It does not establish that:

- the dynamics respect rows and columns;
- interventions factor into two ternary directions;

- admissibility respects the grid;
- or the grid has any preferred physical or semantic significance.

Hence:

$|Z| = mn \not\Rightarrow$ operational factorization $Z \cong Z_m \times Z_n$. (17.7)

A structural factorization requires additional relations.

---

## 17.2 Transverse quotient factorization

A useful exact construction can nevertheless be given.

Suppose there exist two quotient maps

$q_1 : Z \to Z_1$, (17.8)

$q_2 : Z \to Z_2$. (17.9)

Define

$Q(z) = (q_1(z), q_2(z))$. (17.10)

If the two maps jointly separate points,

$q_1(z) = q_1(z')$ and $q_2(z) = q_2(z') \Rightarrow z = z'$, (17.11)

and every coordinate pair is realizable,

$\forall (a, b) \in Z_1 \times Z_2, \exists z \in Z$ such that $q_1(z) = a$ and $q_2(z) = b$, (17.12)

then Q is a bijection.

Therefore,

$Z \cong Z_1 \times Z_2$. (17.13)

This may be called transverse quotient factorization.

The factors are not inferred from cardinality alone. They arise from two compatible families of operational distinctions.

---

## 17.3 A nine-state special case

Suppose

$|Z| = 9$. (17.14)

Let

$P_R = \{R_1, R_2, R_3\}$, (17.15)

$P_C = \{C_1, C_2, C_3\}$ (17.16)

be two partitions of Z, each into three blocks of three states.

If

$|R_i \cap C_j| = 1$ for every i, j, (17.17)

then each state is uniquely identified by one row class and one column class.

Hence

$Z \cong \{1, 2, 3\} \times \{1, 2, 3\}$. (17.18)

This provides a rigorous bridge from nine states to a genuine 3 × 3 incidence structure.

It does not yet produce the LuoShu magic square.

That requires additional balance constraints.

---

# 18. LuoShu as a Parked Extension

The LuoShu branch is potentially relevant to the present framing theory, but it should not be incorporated into the General Framing Kernel before several missing steps are established.

The classical combinatorial part is clear.

Given:

- the values 1 through 9;
- a 3 × 3 grid;
- use of every value exactly once;
- equal sums on all three rows, all three columns, and both main diagonals;

the order-three magic square is unique up to the eight rotations and reflections of the square. Hetu and Luoshu as Semantic Att…

The stronger SMFT claim is different.

It would require deriving the 3 × 3 structure and the balance functional from the dynamics rather than assuming them.

The required chain would be

nine effective states → two transverse ternary quotients → 3 × 3 incidence geometry → balance functional → LuoShu symmetry class. (18.1)

The first nontrivial missing step is therefore not the magic-square calculation.

It is:

> Why should a post-collapse quotient naturally generate two transverse ternary distinctions?

Until this is derived, the LuoShu connection remains a promising extension rather than part of the framing core.

---

# 18.1 What is already mathematically firm

If the 3 × 3 grid and equal-line-sum conditions are imposed, the magic-square uniqueness result is classical.

The stronger claim that this configuration is necessarily the minimum-entropy semantic trace geometry requires an independently justified entropy or stability functional.

The uploaded LuoShu analysis explicitly formulates equal directional sums as the desired balance condition and then interprets the resulting magic square as a minimal-entropy, maximally stable trace structure. Hetu and Luoshu as Semantic Att…

For the present article, these should be separated as follows:

**Theorem:** the combinatorial uniqueness of the 3 × 3 magic square under the imposed constraints.

**Construction or hypothesis:** the claim that semantic post-collapse dynamics naturally generate those constraints.

This distinction prevents a valid combinatorial theorem from being mistaken for a derivation of semantic dynamics.

---

# 19. Naming, Objecthood, and Why They Are Not Yet Part of the Core

One tempting extension is to identify an operational equivalence class

$$[x]_{\sim} \tag{19.1}$$

with an object.

That move is premature.

An equivalence class only states that a collection of microscopic states are currently indistinguishable under a specified protocol.

It does not yet establish:

- semantic identity;
- naming;
- persistence as one object;
- legal or normative identity;
- conceptual categorization;
- or observer-independent objecthood.

Hence:

$$\text{operational equivalence class} \neq \text{Object}. \tag{19.2}$$

Likewise,

$$\text{projection} \neq \text{Naming}. \tag{19.3}$$

The present framework therefore stops before object formation.

---

## 19.1 Why this separation matters

If objecthood were inserted too early, a circularity would arise:

- framing would require knowing what the objects are;
- objecthood would then be explained using the framing that presupposed them.

The present construction avoids this.

The chain

World Formation → Boundary → Quotient → Effective Dynamics (19.4)

is complete enough to operate without presupposing a named ontology.

Objecthood may then be studied as a later construction over already framed distinctions.

---

## 19.2 A future Naming–Objecthood branch

A separate theory may need to explain how framed distinctions become unified into semantic units.

Possible ingredients include:

- naming;
- logical unification;
- rules governing operations on the named unit;
- persistence across frame changes;
- trace stabilization;
- and possibly LuoShu-like post-collapse structures.

These issues are not resolved here.

They should not be silently identified with the quotient construction.

---

# 20. Operational Factorization and Prime-Like Modules

A second parked extension concerns decomposition into irreducible modules.

Suppose an effective system S can be composed from subsystems using a composition rule $\star$:

$S \cong A \star B$. (20.1)

A subsystem P may be called prime relative to $\star$ if

$P \cong A \star B$ (20.2)

implies that one factor is trivial.

The key phrase is “relative to $\star$.”

Different composition grammars may produce different decompositions.

Therefore:

operational primality is composition-relative. (20.3)

This is the appropriate generalization of arithmetic primality.

---

## 20.1 Why prime-number language should be used carefully

For integers,

$p = ab$ (20.4)

with p prime implies

$a = 1$ or $b = 1$. (20.5)

In a general system family,

$P \cong A \star B$ (20.6)

has the same abstract shape.

But general systems need not possess unique prime factorization.

A system may admit more than one inequivalent irreducible decomposition.

Thus the robust conceptual bridge is not

complex systems literally contain prime numbers. (20.7)

It is

prime numbers are one especially clean case of irreducibility under composition. (20.8)

The broader phenomenon is decomposition into operationally irreducible units.

---

## 20.2 Why this branch is not required for the present article

The General Framing Kernel already functions without a unique composition grammar.

It needs only:

- an operational boundary;
- an effective quotient;
- task-relevant dynamics;
- residual testing;
- and recursive revision.

Whether the effective system further decomposes into prime modules is a higher-order structural question.

It is therefore left as an extension.

---

# 21. Composition Grammar and Interface Structure

A related unresolved issue is whether the composition grammar itself can be derived.

If modules A and B interact only through low-dimensional interfaces,

$$I_A, O_A, I_B, O_B, \quad (21.1)$$

one may attempt to define composition through compatible ports.

However, port compatibility alone does not determine the coupling law.

Two compatible systems may be joined through:

- positive feedback;
- negative feedback;
- delayed coupling;
- saturating coupling;
- stochastic coupling;
- or memory-bearing interfaces.

Thus:

$$\text{port compatibility} \not\Rightarrow \text{unique composition semantics.} \quad (21.2)$$

A full theory of system-induced composition would need to derive both interface sufficiency and coupling semantics from operational evidence.

This is important for future work on PORE cells, AI skills, and modular agent architectures, but it is not required for the current framing kernel.

---

# 22. INU and Restoring Coordinates

Nested Uplifts Inevitability provides a related but distinct clue about effective coordinates.

Its Assumption 3.3 considers a deviation coordinate $\Delta$ and a restoring drift $\mu(\Delta)$. In its relaxed engineering form, only local mean-reversion or related sector, passivity, convex-potential, or stochastic conditions are required. Nested Uplifts Inevitability (I...

A representative coarse evolution is

$$d\Delta/d\tau = -h(\Delta) + \eta(\tau). \quad (22.1)$$

If

$$h(0) = 0 \quad (22.2)$$

and

$$h'(0) > 0, \quad (22.3)$$

then $\Delta = 0$ is locally restoring.

The relevance to framing is indirect but important.

A high-dimensional system may possess a low-dimensional restoring coordinate even when the

underlying microstates are much richer.

---

# 22.1 Multiple measurements of one latent factor

The INU analysis considers several possible deviation coordinates, including critical-line offset, whitening gap, and curvature mismatch, all intended to track the same equilibrium through different modalities. Nested Uplifts Inevitability (I…

This illustrates a general caution.

Suppose

$\Delta_1 = f_1(Q)$, (22.4)

$\Delta_2 = f_2(Q)$, (22.5)

$\Delta_3 = f_3(Q)$, (22.6)

where Q is one underlying effective mode.

Then three measured coordinates do not imply three independent factors.

They may be three charts of one latent restoring direction.

Thus:

number of observables ≠ number of effective factors. (22.7)

This distinction will be essential if PORE, LuoShu, or other low-dimensional decompositions are to be tested rigorously.

---

# 23. What Has Been Established

The results of the present development can now be separated by logical status.

---

# 23.1 Core constructions

The following are definitions or structural constructions.

### Minimal World Formation kernel

$K_{abs} = (X, \Lambda, A)$. (23.1)

### Minimal framing kernel

$\Gamma_{min} = (B, \Pi)$. (23.2)

### Operational quotient

$Z = X_B / \sim_\Pi$. (23.3)

### Typed framing residual

$R_F = (r_B, r_D, r_I, r_A, r_M, r_H, \ldots)$. (23.4)

### Self-revising framing loop

$(B, \Pi) \to K \to R \to (B', \Pi')$. (23.5)

These constructions do not assume PORE, LuoShu, objecthood, or unique decomposition.

---

## 23.2 Exact results

Several results follow directly from the definitions.

### Exact quotient dynamics

If equivalence is preserved by all admissible dynamics, the dynamics descend uniquely to the quotient.

### Admission descent

If admissibility is constant on quotient fibers, admission descends uniquely to the effective state space.

### Coarsest exact behavioral quotient

For a fixed boundary, intervention set, observation map, and exact dynamics, behavioral indistinguishability under every finite admissible intervention sequence defines the coarsest exact operational quotient.

These are the strongest mathematical components of the present framework.

---

## 23.3 Conditional propositions

Several results hold only under explicit assumptions.

### Three-dimensional minimality witness

The noisy-confinement example gives rank-three task dependence and therefore establishes local minimal dimension three for that class and task.

### Boundary recursion closure

A composite PORE representation exists when occupancy aggregation, constraint homogenization, agitation scalarization, interface elimination, and operational sufficiency all hold.

### Rank-one environmental drive

A single Heaven–Earth drive is justified when the environmental sensitivity is effectively rank one.

These are conditional results, not universal laws.

---

## 23.4 Engineering hypotheses

The following remain hypotheses requiring empirical testing.

### Boundary Knee Hypothesis

Useful operational boundaries may lie near minima of state complexity plus interface complexity plus residual.

### Broad PORE Closure Hypothesis

Many bounded single-structure systems may admit $\Xi = (\rho, \gamma, \tau)$ as an efficient task-sufficient internal quotient.

### PORE Recursive Closure Hypothesis

Composite systems may often return to the PORE family after appropriate coarse-graining.

These hypotheses are falsifiable and should be treated as such.

---

# 24. Outstanding Problems

The present framework resolves only part of the larger World Formation research program.

The major unresolved questions are as follows.

---

## 24.1 Automatic boundary discovery

The theory currently explains how to evaluate and revise a candidate boundary better than it explains how the first useful boundary is generated.

A future theory should determine whether

$$B^* \tag{24.1}$$

can be recovered from measurable properties such as:

- coupling bottlenecks;
- intervention locality;
- interface compression;
- predictive closure;
- or state–interface complexity trade-offs.

The problem is complicated by the fact that several boundaries may remain simultaneously useful.

---

## 24.2 Approximate quotient mathematics

Exact behavioral equivalence is clean.

Approximate equivalence is not necessarily transitive.

A practical theory therefore needs principled ways to convert noisy similarity structures into operational partitions without pretending that one partition is uniquely forced.

Possible approaches include:

- probabilistic bisimulation;
- rate–distortion formulations;
- information bottlenecks;
- robust lumpability;
- Bayesian model averaging;
- and frame atlases rather than single partitions.

---

## 24.3 Cross-frame translation

If plural frames are fundamental to bounded reasoning, translation becomes a first-class problem.

Given

$\Gamma_A = (B_A, \Pi_A)$, (24.2)

$\Gamma_B = (B_B, \Pi_B)$, (24.3)

one needs to characterize:

- exact translation;
- lossy translation;
- non-invertible translation;
- path-dependent translation;
- and translation holonomy.

The practical question is not whether one frame can replace every other frame, but when knowledge can move reliably between them.

---

## 24.4 Canonical factorization

Even if a quotient Z is well-defined, its decomposition into factors may not be unique.

A future theory must distinguish:

- arbitrary coordinates;
- dynamically invariant factors;
- intervention-typed factors;
- and genuinely canonical decompositions preserved by operational symmetries.

This problem is essential for a stronger justification of PORE coordinates.

---

## 24.5 Why PORE?

The most important outstanding PORE question is no longer simply:

Why three variables?

The stronger question is:

> Why should the three operational roles occupancy, constraint, and agitation recur across multiple domains, observers, and scales?

Evidence would need to show more than dimension three.

A strong result would require:

- rank-three task necessity;
- stable intervention signatures;
- cross-domain recurrence;
- cross-observer recoverability;
- and compositional recurrence.

Only then would $\Xi = (\rho, \gamma, \tau)$ begin to look canonical rather than merely useful.

---

# 24.6 Long memory and trace

Some systems cannot be represented adequately by instantaneous coarse states.

If

$z_t$ (24.4)

does not determine the task-relevant future without knowledge of prior path,

$H_t = \{z_{t-k}, \ldots, z_t\}$, (24.5)

one must decide whether to:

- augment the instantaneous state;
- retain a trace;
- use a non-Markovian kernel;
- split the system into additional modules;
- or reframe the boundary.

This question connects the present framing kernel to the larger World Formation theory of trace, ledger, and time-bearing worlds.

---

# 24.7 Objecthood and Naming

The present article deliberately does not explain how an equivalence class becomes an Object or a Name.

This requires a separate theory of semantic unification.

The outstanding question is not merely whether stable operational distinctions exist, but how such distinctions become grouped, named, governed, and treated as persistent units by an observer.

That problem may involve additional structures not required by the framing kernel.

---

## 24.8 LuoShu emergence

A rigorous LuoShu program requires at least two derivations.

First:

why should a nine-state post-collapse quotient generate two transverse ternary distinctions?

Second:

why should a physically or semantically justified balance functional select the magic-square symmetry class?

Until both questions are answered, the LuoShu connection remains an extension rather than a consequence of the present theory.

---

## 24.9 Composition universality

A major engineering conjecture is that complex practical systems may often be represented not by indefinitely increasing the primitive dimension of one cell, but by composing multiple low-dimensional cells.

This needs quantitative testing.

The comparison should include models of the form

flat d-dimensional state (24.6)

versus

network of N low-dimensional cells. (24.7)

The correct criterion is not variable count alone.

It should include:

- residual;
- interface burden;
- intervention locality;
- transferability;
- reuse;
- and total description complexity.

---

# 25. A Research Program

The framework is sufficiently explicit to support controlled experiments.

A useful research program can be organized into four benchmark families.

---

## 25.1 Benchmark A — Exact finite-state recovery

Construct finite controlled systems with known behavioral equivalence classes.

Provide the learner with:

- observations;
- intervention access;
- hidden microscopic state;
- and a defined task output.

Test whether the learner recovers

$$Z_P^* = X_B / \equiv_P. \quad (25.1)$$

Metrics should include:

- quotient accuracy;
- number of probes;
- intervention efficiency;
- and convergence to the exact partition.

This benchmark tests the mathematical core directly.

---

## 25.2 Benchmark B — Bounded noisy framing

Add:

- observation noise;
- limited probing depth;
- missing interventions;
- and finite tolerance.

Now exact equivalence is no longer directly observable.

Measure:

- number of admissible partitions;
- stability of partitions across samples;
- sensitivity to tolerance;
- and whether new probes correctly refine rather than arbitrarily replace previous partitions.

This benchmark tests bounded framing.

---

## 25.3 Benchmark C — PORE dimensional competition

Choose systems for which candidate coarse models of dimensions

$$d = 1, 2, 3, 4, \ldots \quad (25.2)$$

can be fit.

Compare

$$J_d = \delta_d + \lambda C_d, \quad (25.3)$$

where $\delta_d$ measures task residual and $C_d$ model complexity.

The central empirical questions are:

- Does d = 3 repeatedly occupy the complexity–performance knee?
- Are the three dimensions recoverable as occupancy-, constraint-, and agitation-like roles?
- Do intervention signatures distinguish those roles?
- Does the third direction remain necessary outside narrow training conditions?

This benchmark directly tests the PORE³ hypothesis.

---

## 25.4 Benchmark D — Composition and re-coarse-graining

Construct networks of low-dimensional cells.

Compose

$C_1, C_2, \ldots, C_N$ (25.4)

under controlled coupling strengths, timescale separations, and noise structures.

Then ask whether a macro compiler exists,

$C_{macro} : X_{full} \to \Xi^*$. (25.5)

Measure closure defects

$\delta_D, \delta_M, \delta_I, \delta_A$. (25.6)

Vary:

- coupling strength;
- interface timescale;
- anisotropy;
- covariance;
- number of stocks;
- environmental rank.

The aim is to map the region in which the PORE family is approximately closed under composition.

---

# 26. Implications for Artificial Intelligence

The immediate AI implication is not that an AGI should be programmed with three PORE variables everywhere.

The deeper implication concerns framing.

Current AI systems often receive a representation chosen in advance.

Variables, tools, documents, agents, memory structures, and task boundaries are frequently specified externally.

A stronger agent would need to revise those representational decisions.

---

# 26.1 From World Models to World-Framing Engines

A conventional world model asks:

Given a state representation z, what happens next? (26.1)

A framing engine must ask a prior question:

What should count as z? (26.2)

The framing problem therefore includes:

- choosing a boundary;
- quotienting irrelevant distinctions;
- detecting when the quotient is too coarse;
- revising the boundary;
- and maintaining translations between alternative frames.

A self-revising AI architecture would therefore contain something like

raw experience → candidate frame → effective dynamics → active probe → residual → revised frame. (26.3)

This is qualitatively different from merely fitting a larger predictor.

---

# 26.2 Active framing

Suppose two candidate frames $\Gamma_A$ and $\Gamma_B$ make similar predictions under passive observation.

The agent can select an intervention $u^*$ that maximally separates their predicted outcomes.

Schematically,

$u^* = \arg\max_u D(P_A(Y \mid u), P_B(Y \mid u))$. (26.4)

The result of the experiment updates the admissibility of the candidate frames.

Thus framing can become an active scientific process.

The agent does not merely update beliefs inside a representation.

It actively tests the representation itself.

---

# 26.3 PORE as a first internal compiler

Within such an architecture, PORE may have a practical role even if its strongest theoretical claims remain unproven.

When a bounded single-structure boundary has been provisionally identified, the system can first test

$\Pi_{PORE} : X_I \to (\rho, \gamma, \tau)$. (26.5)

If closure succeeds, the triple provides a compact control representation.

If closure fails, the typed residual determines whether to:

- add an environmental mode;
- refine occupancy;
- tensorize constraint;
- retain memory;
- split the system;
- or redraw the boundary.

PORE therefore becomes a first-pass compiler rather than a dogmatic ontology.

---

# 27. Implications for the Wider World Formation Program

The present development also clarifies several earlier strands of World Formation theory.

---

## 27.1 From admission to framing

The original admission operator answers

Which states pass the boundary condition? (27.1)

The framing quotient answers

Which distinctions must survive after admission? (27.2)

These are complementary but different operations.

---

## 27.2 From framing to recursive revision

Once a frame is constructed, residual exposes the limits of that frame.

The loop

$\Gamma \rightarrow R \rightarrow \Gamma'$ (27.3)

therefore supplies a concrete mechanism for self-revision.

This connects naturally to the broader World Formation architecture in which admissibility rules, declarations, and worlds can themselves become objects of later revision.

---

## 27.3 From static worlds to time-bearing worlds

The present quotient theory is compatible with both memoryless and memory-bearing systems.

When instantaneous projection fails because history matters, the state may need to be lifted from

z_t (27.4)

to

(z_t, H_t). (27.5)

This marks a transition from static framing toward trace-bearing and time-bearing worlds.

The framing kernel therefore does not replace the earlier ledger and recursive-time constructions.

It supplies a lower-level representation problem that those later structures can inherit.

---

# 28. Scope and Non-Claims

To prevent the framework from being interpreted more strongly than the results warrant, several non-claims should be stated explicitly.

The present work does not prove that:

1. reality has one unique decomposition;
2. every system has a unique natural boundary;
3. every useful quotient is three-dimensional;
4. PORE variables are fundamental physical coordinates;
5. every composite PORE system returns to a PORE triple;
6. nine effective states necessarily form LuoShu;
7. equivalence classes are Objects;
8. operational irreducibility implies literal arithmetic primality;
9. frame stability implies metaphysical truth;
10. different cultures or observers must eventually converge to one identical representation.

Instead, the framework proposes a way to state these stronger questions precisely enough to test them.

That is already substantial progress.

---

# 29. Summary of the General Framing Kernel

The main architecture can now be stated compactly.

### Step 1 — World Formation

Introduce nontrivial admissibility:

A : X → {0, 1}. (29.1)

### Step 2 — Operational boundary

Choose or generate

B. (29.2)

### Step 3 — Quotient

Construct

$\Pi : X_B \to Z$. (29.3)

### Step 4 — Effective dynamics

Test whether

$\Pi \circ F_u \approx K_u \circ \Pi$. (29.4)

### Step 5 — Audit

Compute

$R_F = (r_B, r_D, r_I, r_A, r_M, r_H, \ldots)$. (29.5)

### Step 6 — Repair or reframe

If the residual is parameter-level,

$\theta \to \theta'$. (29.6)

If the quotient is too coarse,

$\Pi \to \Pi'$. (29.7)

If the boundary is wrong,

$B \to B'$. (29.8)

### Step 7 — Repeat

$\Gamma_{n+1} = F_{frame}(\Gamma_n, R_n, E_n)$. (29.9)

This is the Self-Revising Framing Kernel.

---

# 30. The Single-Structure Working Protocol

For many practical human and AI analyses, the operational boundary is already approximately known.

In that case the full general search can be replaced by a simpler working protocol:

$B \to \Delta_E \to (\rho, \gamma, \tau) \to \eta \to R$. (30.1)

The steps are:

1. provisionally fix the boundary B;
2. extract the dominant environmental drive $\Delta_E$;
3. attempt the PORE internal quotient $\Xi = (\rho, \gamma, \tau)$;
4. retain secondary environmental modes $\eta$ when necessary;
5. audit the residual;
6. repair or reframe if closure fails.

This protocol should be regarded as a high-value special case of the General Framing Kernel, not

as its definition.

---

# 31. Conclusion

The problem addressed in this article is more primitive than ordinary model fitting.

Before an observer can predict a system, the observer must already possess some answer to two questions:

What is currently inside the problem? (31.1)

Which differences inside it are still worth distinguishing? (31.2)

The proposed minimal answer is

$\Gamma_min = (B, \Pi)$. (31.3)

The boundary B determines the operational inside–outside cut.

The projection Π determines which microscopic distinctions survive into the effective state space.

From this simple starting point, several results follow.

For a fixed exact protocol, states that cannot be distinguished by any finite admissible intervention sequence form a behavioral equivalence relation. Its quotient is the coarsest exact operational frame.

For a bounded observer, however, this exact quotient is inaccessible. Finite probing depth, noise, limited interventions, finite resolution, and tolerance produce provisional equivalences. Learning may therefore split previously merged states, while deeper failures may force either quotient refinement or boundary migration.

The resulting picture is neither a theory of one mandatory decomposition nor an unrestricted relativism.

Multiple frames may remain valid.

Their quality can still be compared through closure, prediction, intervention, admissibility, robustness, complexity, and translation.

The General Framing Kernel can therefore be summarized as

Boundary + Quotient + Dynamics + Residual + Revision. (31.4)

Within this wider architecture, the Heaven–Earth plus PORE construction becomes a concrete special hypothesis.

A boundary-relative environmental sensitivity may possess a dominant rank-one direction,

$\Delta_E = v_1^T e$, (31.5)

while the internal dynamics may admit a three-role control quotient,

$\Xi = (\rho, \gamma, \tau)$. (31.6)

The current evidence justifies treating this as a testable low-dimensional control architecture,

not as universal ontology.

The strongest version of the future program is therefore not:

Reality is fundamentally three-dimensional in PORE variables. (31.7)

Nor is it:

There exists one objectively privileged decomposition of every system. (31.8)

The stronger and more defensible research question is:

> Under what classes of dynamics, tasks, interventions, and scales do bounded observers repeatedly recover the same low-dimensional operational structures, and when do those structures survive composition, translation, and revision?

That question is experimentally accessible.

It also leaves room for the deeper unresolved branches: Naming, Object formation, LuoShu trace geometry, prime-like decomposition, emergent composition grammar, and the relation between framed operational worlds and richer semantic worlds.

The present result is therefore best understood as a kernel rather than a completed ontology.

World Formation supplies the logic of admissibility.

Boundary supplies the operative domain.

Quotient supplies the retained distinctions.

Dynamics tests whether those distinctions close.

Residual reveals where the frame fails.

Revision allows the frame itself to change.

Together they define a constructive route by which a bounded observer can form, test, revise, and translate operational worlds without assuming that the world arrives already partitioned.

# Appendix A. Formal Notation and Logical Status

This appendix collects the main symbols used throughout the article and separates definitions from stronger claims.

## A.1 Core spaces

Let

$X$ (A.1)

denote the microscopic or pre-framed state space.

A provisional operational boundary $B$ selects an internal domain

$X_B \subseteq X$. (A.2)

The environment relative to $B$ is denoted

E_B. (A.3)

A projection or quotient map is

Π : X_B → Z. (A.4)

The induced equivalence relation is

x ∼_Π y ⇔ Π(x) = Π(y). (A.5)

Hence

Z ≅ X_B / ∼_Π. (A.6)

The minimal frame is

Γ_min = (B, Π). (A.7)

---

# A.2 Dynamic extension

For admissible intervention u,

F_u : X_B → X_B (A.8)

denotes the microscopic dynamics.

If the quotient is dynamically closed, there exists

K_u : Z → Z (A.9)

such that

Π ∘ F_u = K_u ∘ Π. (A.10)

The effective dynamic frame is therefore

Γ_dyn = (B, Π, K). (A.11)

---

# A.3 Measurement and admissibility

Let

M : X_B → Y_M (A.12)

denote task-relevant measurement.

Let

A_X : X_B → {0, 1} (A.13)

denote microscopic admissibility.

If both descend to the quotient, then there exist

Mˉ : Z → Y_M (A.14)

and

$A_Z : Z \to \{0, 1\}$ (A.15)

such that

$M = M' \circ \Pi$ (A.16)

and

$A_X = A_Z \circ \Pi$. (A.17)

---

# A.4 Behavioral equivalence

For an admissible intervention word

$w = (u_1, \ldots, u_n)$, (A.18)

define

$F_w = F_{u_n} \circ \ldots \circ F_{u_1}$. (A.19)

Given task-relevant output

$o : X_B \to Y$, (A.20)

define exact behavioral equivalence by

$x \equiv_P y \Leftrightarrow o(F_w(x)) = o(F_w(y))$ for every finite admissible $w$. (A.21)

The coarsest exact operational quotient is

$Z_{P^*} = X_B / \equiv_P$. (A.22)

---

# A.5 Bounded-observer quotient

For finite probing depth $H$,

$x \equiv_H y \Leftrightarrow o(F_w(x)) = o(F_w(y))$ for every $|w| \leq H$. (A.23)

The associated quotient is

$Z_H = X_B / \equiv_H$. (A.24)

The filtration is

$\equiv_0 \supseteq \equiv_1 \supseteq \equiv_2 \supseteq \ldots \supseteq \equiv_\infty$. (A.25)

---

# A.6 Framing residual

A useful typed residual is

$R_F = (r_B, r_D, r_I, r_A, r_M, r_H, \ldots)$. (A.26)

where:

- r_B measures boundary inadequacy;
- r_D measures dynamic nonclosure;
- r_I measures intervention inconsistency;
- r_A measures admissibility inconsistency;
- r_M measures measurement insufficiency;
- r_H measures hidden-history or memory failure.

A candidate frame is admitted relative to tolerance ε_F when

$R_F(\Gamma) \leq \varepsilon_F$. (A.27)

---

# Appendix B. Theorem, Proposition, Construction, and Hypothesis Register

The framework contains claims of different strength. They should not be conflated.

# B.1 Theorem-level results

### Theorem B.1 — Exact quotient dynamics

If

$\Pi(x) = \Pi(y) \Rightarrow \Pi(F_u(x)) = \Pi(F_u(y))$ (B.1)

for every admissible u, then a unique quotient dynamics K_u exists satisfying

$\Pi \circ F_u = K_u \circ \Pi$. (B.2)

---

### Theorem B.2 — Admission descent

If

$\Pi(x) = \Pi(y) \Rightarrow A_X(x) = A_X(y)$, (B.3)

then a unique quotient-level admission map A_Z exists such that

$A_X = A_Z \circ \Pi$. (B.4)

---

### Theorem B.3 — Coarsest exact behavioral quotient

For fixed B, U, F, and o, exact behavioral indistinguishability under all finite admissible intervention sequences defines the coarsest exact operational quotient.

This is one of the strongest results in the present article.

---

# B.2 Conditional propositions

### Proposition B.1 — Rank-one environmental reduction

If the environmental sensitivity matrix

$J_E = \partial F / \partial e$ (B.5)

has a dominant singular value

$\sigma_1 \gg \sigma_2$, (B.6)

then the leading environmental influence is approximately represented by

$\Delta_E = v_1^T e$. (B.7)

This is a local low-rank result, not a universal Heaven–Earth theorem.

---

### Proposition B.2 — Three-dimensional minimality witness

If the task outputs depend locally with rank three on

$\Xi = (\rho, \gamma, \tau)$, (B.8)

then no exact smooth representation of local dimension less than three can preserve those outputs.

The noisy-confinement example provides one explicit witness.

---

### Proposition B.3 — Recursive PORE closure

If BR1–BR5 hold—

- occupancy aggregation;
- constraint homogenization;
- agitation scalarization;
- fast interface or finite memory;
- operational sufficiency—

then the composite admits an approximate macro PORE representation.

---

### Proposition B.4 — Finite strict-descent termination

If the candidate frame set is finite and every accepted update strictly lowers one common objective J, then the revision sequence terminates in finitely many steps at a local minimum.

This does not imply global optimality.

---

# Appendix C. No-Go Results

A central purpose of the development was to identify what cannot be inferred from weaker premises.

## C.1 Static filtering does not generate history

Commuting static admission filters cannot by themselves produce path dependence or a time arrow.

---

## C.2 Arbitrary disclosure does not guarantee a clock

An event algebra need not possess a consistent additive grading.

Recursive depth and physical time must therefore be distinguished.

---

## C.3 Pairwise consistency does not imply higher-order consistency

A pairwise-complete grammar may still fail at triadic or higher arity.

---

## C.4 Fixed-frame repair cannot produce genuine reframing

If the repair operator is restricted to one frame codomain, it cannot generate a new frame.

---

## C.5 Search failure does not prove absence of repair

Failure to find a within-frame repair may reflect limited search rather than genuine impossibility.

---

## C.6 Finite trace does not identify the full repair envelope

Observed repair possibilities form only a lower bound on the true repair space.

---

## C.7 Composition does not automatically preserve SS or PORE form

Two three-coordinate systems generally produce a six-dimensional composite before further assumptions are imposed.

---

## C.8 Child minimality does not imply parent minimality

Even if each subsystem requires three effective coordinates, the composite may require fewer, equal, or more.

---

## C.9 Compression does not imply dynamical closure

A low-dimensional representation may reconstruct or cluster data well while failing to support autonomous effective dynamics.

---

## C.10 Approximate indistinguishability need not be transitive

Tolerance-based similarity does not automatically define a valid quotient.

---

## C.11 A frame does not require an object ontology

The pair

$\Gamma_{min} = (B, \Pi)$ (C.1)

is meaningful without assigning Objects or Names to equivalence classes.

---

## C.12 Boundary alone is not a frame

A cut determines inside and outside but does not specify which internal distinctions survive.

At least

$(B, \Pi)$ (C.2)

is required.

---

## C.13 External behavior does not uniquely identify a physical boundary

Different microscopic cuts may realize the same effective transducer.

---

## C.14 Effective dimension does not determine factor roles

$\dim Z = 3$ (C.3)

does not imply three privileged coordinates, much less the specific roles $\rho$, $\gamma$, and $\tau$.

---

## C.15 Nine states do not imply LuoShu

$|Z| = 9$ (C.4)

does not imply

$Z \cong 3 \times 3$ (C.5)

in any operationally meaningful sense.

A structural bridge must be separately derived.

---

## C.16 Three-dimensional PORE failure does not imply PORE⁴

An additional degree of freedom may represent:

- memory;
- anisotropy;
- a second stock;
- an environmental mode;
- another subsystem;
- or a boundary error.

Hence:

$d_{eff} > 3 \not\Rightarrow$ fourth primitive PORE coordinate. (C.6)

---

## C.17 Operational equivalence does not imply Objecthood

$[x]_{\sim}$ (C.7)

is not yet a Name, Object, identity, or ontological unit.

---

## C.18 Irreducibility depends on composition grammar

A system may be prime relative to one composition rule and decomposable under another.

Operational primality is therefore grammar-relative.

---

# Appendix D. The Practical Single-Structure Protocol

For many practical applications, the general framing problem is unnecessarily broad because an approximate boundary is already available.

In such cases, a compact protocol can be used.

## D.1 Step 1 — Provisional boundary

Choose

$B_0$. (D.1)

This is not asserted to be the uniquely correct boundary.

It is a working cut.

---

## D.2 Step 2 — Principal environmental drive

Estimate

$J_E = \partial F / \partial e$. (D.2)

If

$\sigma_1 \gg \sigma_2$, (D.3)

define

$\Delta_E = v_1^T e$. (D.4)

Interpret the two ends of this scalar drive as opposite environmental anchors.

---

## D.3 Step 3 — Internal PORE compiler

Attempt

$\Pi_I : X_I \to (\rho, \gamma, \tau)$. (D.5)

Use functional definitions:

$\rho$ = effective occupancy or content; (D.6)

$\gamma$ = effective constraint or holding strength; (D.7)

$\tau$ = effective agitation or coherence loss. (D.8)

---

## D.4 Step 4 — Secondary environment

Retain residual environmental modes

$\eta = (\eta_2, \eta_3, \ldots)$ (D.9)

only when their omission materially increases task residual.

---

## D.5 Step 5 — Closure audit

Estimate

$R_{SS} = (r_D, r_I, r_A, r_M, r_H, r_E)$. (D.10)

where $r_E$ measures failure of the rank-one environmental approximation.

---

## D.6 Step 6 — Repair route

If the residual is parameter-level:

repair θ. (D.11)

If occupancy is insufficient:

refine ρ. (D.12)

If constraint is anisotropic:

$\gamma \to \vec{\gamma}$ or Γ. (D.13)

If agitation hides long memory:

$\tau \to (\tau, H)$ or a non-Markovian model. (D.14)

If the environment has rank greater than one:

$\Delta_E \to (\Delta_1, \Delta_2, \ldots)$. (D.15)

If a new autonomous loop appears:

consider a multi-cell representation. (D.16)

If the whole cut remains unstable:

$B \to B'$. (D.17)

This gives the practical cycle

$B \to \Delta_E \to (\rho, \gamma, \tau) \to \eta \to R \to$ repair/reframe. (D.18)

---

# Appendix E. A Minimal Blind Numerical Experiment

A direct empirical program can now be formulated without assuming that PORE is correct.

## E.1 Ground-truth generator

Choose a controlled dynamical system

$x_\{t+1\} = F(x_t, u_t) + \xi_t$. (E.1)

Hide the microscopic state decomposition from the learner.

Expose only:

- task outputs;
- permitted interventions;
- measurement noise;
- training trajectories.

---

## E.2 Competing representations

Fit models with effective dimensions

$d = 1, 2, 3, 4, 5, \ldots$ . (E.2)

For each d, optimize

$J_d = R_d + \lambda C_d$. (E.3)

$R_d$ should include at least:

- predictive error;
- intervention error;
- admissibility error;
- held-out regime error.

---

# E.3 PORE-specific test

For $d = 3$, do not merely fit arbitrary latent coordinates.

Test whether the three recovered directions can be assigned stable operational signatures corresponding to:

- occupancy;
- constraint;
- agitation.

Let the relevant intervention-response matrix be

$G = \partial z / \partial u$. (E.4)

A stronger PORE result requires that, after permissible reparameterization, G exhibit stable role differentiation rather than arbitrary coordinate mixing.

---

# E.4 Blindness condition

The analyst evaluating the model should not know which candidate dimensionality generated the data.

Likewise, the model-selection procedure should not be told that $d = 3$ is preferred.

This prevents the benchmark from being engineered to recover PORE.

---

# E.5 Success criteria

A meaningful positive result would require several of the following:

1. $d = 3$ lies near the prediction–complexity knee;
2. the third singular direction remains above task tolerance;
3. the recovered axes show stable intervention roles;
4. the same roles recur across independent systems;
5. the roles survive composition and coarse-graining;
6. equivalent results are obtained from different initializations and observers.

Only this combination would provide evidence for a recurrent PORE structure.

# Appendix F. Boundary Recursion Stress Tests

The PORE recursion hypothesis should be attacked with minimal counterexamples rather than confirmed only on favorable examples.

## F.1 Two-stock stress test

Use

$$\dot{a} = u_a - \alpha a, \quad \text{(F.1)}$$

$$\dot{b} = u_b - \beta b. \quad \text{(F.2)}$$

Test whether a scalar aggregate

$$\rho = a + b \quad \text{(F.3)}$$

remains sufficient as $\alpha - \beta$ increases.

The expected transition is:

single-stock closure → approximate failure → two-stock representation. (F.4)

---

## F.2 Anisotropy stress test

Use

$$d\mathbf{x} = -K\mathbf{x}\, dt + \sqrt{(2\tau)}\, d\mathbf{W}, \quad \text{(F.5)}$$

with

$$K = \mathrm{diag}(\gamma_x, \gamma_y). \quad \text{(F.6)}$$

Increase

$$|\gamma_x - \gamma_y|. \quad \text{(F.7)}$$

Measure when a scalar $\gamma$ ceases to preserve intervention response.

---

## F.3 Memory stress test

Use

$$\dot{x} = -\gamma x + \eta, \quad \text{(F.8)}$$

$$d\eta = -\eta/\theta\, dt + \sigma\, dW. \quad \text{(F.9)}$$

Increase $\theta$ relative to the macro timescale.

This tests the transition

Markovian triple → hidden-memory failure → history lift. (F.10)

---

# F.4 Environmental-rank stress test

Construct

$\dot{p} = Ap + Be.$ (F.11)

Vary the singular spectrum of B.

Track performance as

$\sigma_2/\sigma_1$ (F.12)

increases.

This directly tests the Heaven–Earth rank-one approximation.

---

# F.5 Composition stress test

Couple two PORE-like cells:

$\dot{x}_A = f_A(x_A) + \kappa C_{AB} x_B,$ (F.13)

$\dot{x}_B = f_B(x_B) + \kappa C_{BA} x_A.$ (F.14)

Sweep $\kappa$ from weak coupling to strong synchronization.

Measure whether the preferred macro representation transitions through regimes such as

two separate triples → coupled multi-cell model → one macro triple. (F.15)

Such a result would be particularly informative because it would show that the appropriate framing itself changes with coupling regime.

---

# Appendix G. Outstanding Research Roadmap

The unresolved problems can be organized by logical dependence rather than by speculative importance.

# G.1 Tier 1 — Core framing mathematics

These should be addressed first.

## G1. Approximate behavioral quotient

Develop a mathematically stable substitute for exact equivalence under noise and tolerance.

## G2. Boundary–quotient co-optimization

Study how changes in B alter the minimal sufficient $\Pi$.

## G3. Active discrimination

Develop probe-selection rules that efficiently split false equivalence classes.

### G4. Cross-frame translation

Formalize exact, approximate, and path-dependent translation between operational quotients.

These are direct extensions of the article's central kernel.

---

## G.2 Tier 2 — PORE validation

### P1. Recoverability

Can ρ, γ, and τ be inferred from raw data without manually naming them?

### P2. Typed intervention signatures

Do the three roles remain distinguishable under controlled perturbation?

### P3. Dimensional competition

Does three repeatedly outperform two without being dominated by four?

### P4. Composition recurrence

Does coarse-graining of multi-cell systems regenerate the same three-role structure?

### P5. Failure taxonomy

Can observed failures be reliably classified as stock, anisotropy, memory, environmental rank, or boundary errors?

A successful PORE program should answer these empirically.

---

## G.3 Tier 3 — Framing plurality and cultural persistence

A major conceptual consequence of the framework is that a previously formed decomposition need not become useless merely because a different environment supports another decomposition.

Future work should study conditions under which a transported frame remains effective.

Let

Γ_A (G.1)

be a frame learned in environment A and

Γ_B (G.2)

a locally developed frame in environment B.

One may compare:

Performance(Γ_A | B) (G.3)

with

Performance(Γ_B | B). (G.4)

The interesting cases are not only those where one dominates.

There may be complementary frames whose joint use reveals distinctions invisible to either alone.

This suggests that cultural or disciplinary plurality may sometimes function as a source of alternative coarse-grainings rather than merely as noise around one optimal representation.

That claim remains a hypothesis, but it follows naturally from the non-uniqueness of bounded operational framing.

---

# G.4 Tier 4 — Naming and Objecthood

This branch should begin only after the framing kernel is stable.

Its central problem is:

How does a set of effective distinctions become unified into a named semantic unit? (G.5)

Questions include:

- What makes an equivalence class eligible for Naming?
- How is one Name maintained across frame changes?
- How do operations on the Name acquire law-like constraints?
- How does persistence differ from mere state similarity?
- Can a Name survive translation between non-isomorphic frames?

This is where the separate theory of 名、道、法 and logical unification may become relevant.

The present article deliberately leaves these questions open.

---

# G.5 Tier 5 — LuoShu and post-collapse structure

The minimal research target is

|Z| = 9 (G.6)

plus two operationally derived ternary partitions satisfying

|R_i ∩ C_j| = 1. (G.7)

Only then is

Z ≅ 3 × 3 (G.8)

structurally justified.

A second theorem would then need to derive a balance functional whose minima are precisely the LuoShu symmetry orbit.

The uploaded LuoShu analysis already establishes the combinatorial uniqueness of the 3 × 3 magic square under the imposed equal-line-sum conditions, but deriving those conditions from semantic or collapse dynamics remains outstanding. Hetu and Luoshu as Semantic Att…

---

# G.6 Tier 6 — Operational prime decomposition

The long-term question is whether some system classes admit decomposition into operationally irreducible modules relative to a system-supported composition grammar.

The key difficulty is not merely defining irreducibility.

It is deriving the relevant composition grammar rather than imposing it.

This should remain a later branch.

---

# Appendix H. Relation to the Wider World Formation Development

The present article occupies a specific position in the broader theoretical sequence.

A useful high-level chain is

Assumption → Operator → Filtration → Declaration → Self-Revision → Time-Bearing World. (H.1)

The framing kernel developed here can be inserted between primitive World Formation and richer world construction.

A refined schematic is

Admissibility → Boundary → Projection → Effective Dynamics → Residual → Declaration/ Revision → Trace → Time-Bearing World. (H.2)

The contribution of the present article is concentrated in the middle:

Boundary → Projection → Effective Dynamics → Residual → Reframing. (H.3)

It therefore does not replace the earlier work on declaration, trace, recursive depth, or time.

It supplies a missing account of how a bounded observer can obtain the effective state space on which those later operations act.

---

# H.1 Relation to PORE

The Post-Ontological Reality Engine is especially compatible with this framing architecture because ontology need not be fixed in advance.

The sequence becomes

raw dynamics → operational framing → provisional effective states → recursive revision. (H.4)

Ontology, if later required, can emerge downstream rather than being assumed as primitive input.

This is arguably a more precise sense in which the architecture is “post-ontological.”

---

## H.2 Relation to recursive disclosure

A frame determines what distinctions are currently available.

Disclosure may reveal distinctions that were previously quotient-equivalent.

Thus recursive disclosure can refine

$\Pi_n \rightarrow \Pi_{n+1}$. (H.5)

The filtration of behavioral equivalence classes therefore gives one concrete operational realization of progressive disclosure.

---

## H.3 Relation to Declaration

When several approximate frames remain admissible,

$\Omega_F = \{\Gamma_1, \Gamma_2, \ldots\}$, (H.6)

Declaration selects one operative frame when action requires commitment.

This does not imply that the rejected alternatives become false.

They may remain stored in the frame atlas.

Declaration therefore functions as operational commitment rather than metaphysical annihilation of alternatives.

---

# Appendix I. Compact Statement of the Present Theory

The whole article can be compressed into the following sequence.

A bounded observer begins with a domain X.

A nontrivial World Formation criterion defines admissibility:

$A : X \rightarrow \{0, 1\}$. (I.1)

An operational cut selects a working inside:

$B : X \rightarrow X_B$. (I.2)

A projection identifies currently irrelevant distinctions:

$\Pi : X_B \rightarrow Z$. (I.3)

The frame is

$\Gamma = (B, \Pi)$. (I.4)

If the projection is dynamically congruent, microscopic dynamics descend:

$\Pi \circ F_u = K_u \circ \Pi$. (I.5)

The frame is audited through typed residual:

$R_F = (r_B, r_D, r_I, r_A, r_M, r_H, \ldots)$. (I.6)

Failure may produce:

parameter repair, (I.7)

quotient refinement, (I.8)

boundary migration, (I.9)

history lift, (I.10)

or multi-cell decomposition. (I.11)

For bounded single-structure systems, a useful special proposal is

$\Gamma_{SS} = (B, \Delta_E, \rho, \gamma, \tau, \eta, K)$. (I.12)

where

$\Delta_E$ (I.13)

is a boundary-relative dominant environmental drive and

$\Xi = (\rho, \gamma, \tau)$ (I.14)

is a candidate minimal internal control quotient.

The resulting architecture is not a fixed ontology.

It is a recursive mechanism for constructing and revising operational worlds.

---

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14. $\mathbb{O} \to \mathbf{G_2/SO(4)} \to \mathbb{H} \to \mathbb{C}^2$ 成界過程初探 **1–25.**
15. 遞歸開顯、先天八卦、漫話成界之學**.**
16. 以「成界之學」攻剋「先天八卦」本義**.**

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# Closing Research Statement

The present theory should therefore be judged by a relatively narrow standard.

It need not yet solve ontology, Naming, LuoShu, consciousness, or universal system decomposition.

Its first obligation is simpler:

> Given a bounded observer, a set of admissible interventions, and a task, can the framework correctly determine which distinctions may be removed, detect when too much has been removed, and revise the boundary or quotient when evidence demands it?

If the answer is yes, then the General Framing Kernel already performs useful theoretical work.

If PORE³ subsequently proves to recur across independently constructed systems, survive blind dimensional competition, exhibit stable typed intervention signatures, and reappear after composition, then a stronger claim becomes justified.

If it does not, the framing kernel remains intact.

That asymmetry is important.

It makes the architecture falsifiable in its special claims without making the entire theory depend on any one preferred decomposition.

The present development therefore ends with a deliberately layered position:

**World Formation is the logic of admissibility.**

**Framing is the logic of operational distinction.**

**Quotient is the logic of safe unification.**

**Dynamics is the test of whether the unification closes.**

**Residual is the evidence that the current frame has reached its limit.**

**Revision is the mechanism by which a bounded observer constructs a new operational world.**

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

*I am merely a midwife of knowledge.*